

Statistical Modeling

David Puelz

The University of Austin



Course goals

- Use supervised learning to build models for prediction and for understanding (causal inference)
- Understand the benefits and limitations of the models we build
- Given a new situation, select an appropriate model, carry it out, and effectively communicate the results
- This course focuses on practice and implementation. But ... we will pick up theoretical details when necessary.



Who am I?

- **Educator:** Teaching at UATX since 2024, UT Austin from 2020-2024, graduate student from 2013-2018. Previously at the University of Chicago Booth School of Business and Goldman Sachs.
- **Researcher:** I develop new statistical/machine learning methods for analyzing complex data, with applications in policy, economics, and the social sciences.
- **Human:** Grew up in Dallas. Husband and dad with 2 girls + 1 boy + 1 dog. I like working out, smoking meat, golf, and cycling.



Introduction to the course

Course logistics

Supervised learning

Building intuition: k-nearest neighbors

The bias-variance tradeoff

Answer two questions for me:

What is one thing you're *excited* about for this class?

What is one thing you're *nervous or concerned* about for this class?

Pass around your answers. Another student will read your responses, so everyone stays anonymous.

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Populi & Github

- Access at `uaustin.populiweb.com`
- Access at `https://github.com/dpuelz/StatisticalModeling`
- Github pages will contain `schedule`, `homeworks`, `data`, `code`, ...
- **Github will be our main landing page.** Populi will be used for grades and administrative tasks.



Class structure

- Understanding the concepts really only comes from practice



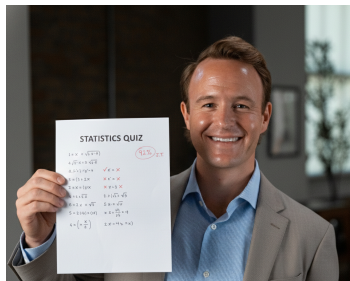
Class structure

- Understanding the concepts really only comes from practice
- Class time will be mostly lecture and discussion. We will also have “in-class reps” on Quiz reviews and exercises.



Quizzes

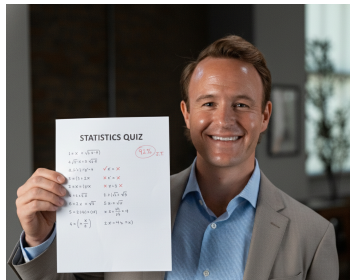
- 5 quizzes, taken during class on the 2nd, 4th, 6th, 8th, and 10th Friday



-
- A man in a light blue shirt and grey blazer is smiling and holding a white sign that reads "STATISTICS QUIZ". The sign lists several mathematical formulas, including the quadratic formula, binomial expansion, area of a circle, and various trigonometric and logarithmic identities. A red circle highlights the formula $\frac{1}{2} \pi r^2$ on the sign.

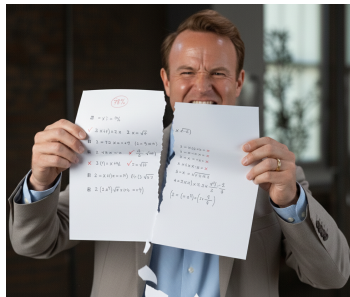
Quizzes

- 5 quizzes, taken during class on the 2nd, 4th, 6th, 8th, and 10th Friday
- Assessing your knowledge of lecture material up to that point
- Concepts assessed will come directly from the homeworks



Homework

- Why homework?

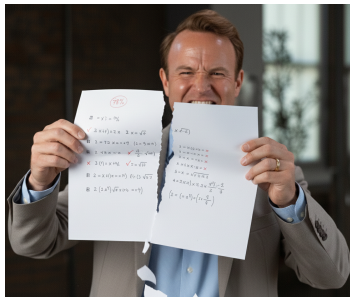


- [illegible]



Homework

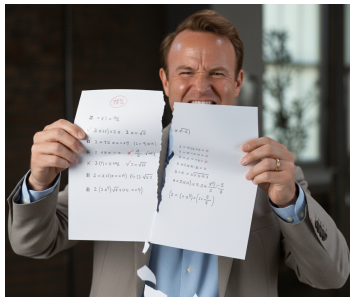
- Why homework?
- 5 homework assignments, due on the 2nd, 4th, 6th, 8th, and 10th Friday (like the quizzes)
- **Homeworks can be completed individually or in groups of up to three people.** They must be turned in by the start of each Friday class. Names of all group members must be included on the writeups.





Homework

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- **Homeworks can be completed individually or in groups of up to three people.** They must be turned in by the start of each Friday class. Names of all group members must be included on the writeups.
- I expect professional writeups in a markdown format (Rmd) or (Jupyter)





A pedagogical note on AI

LLMs can be useful for the homeworks and learning concepts. I encourage their use! And ... I will not police their use. I only ask that you cite when LLMs are used.

However, if overused & relied on too heavily, **you will not learn anything**.

The quizzes are designed from the homeworks and will assess how well you understand the concepts. **Through the quizzes, you will get feedback quickly on whether you've overused AI!** Use this feedback to optimize your learning with AI.



- The exam will be during the 11th week (final exam week). There are no make-ups or alternative dates, so please plan accordingly!
- It will be a free response, written exam taken with pen and paper (exactly like our quizzes)



Grading

Homework	20%
Exam	30%
Quizzes	50%



Statistical computing

- We will use **R** for statistical analysis throughout the course
- You are definitely welcome to use **Python**, but the course materials will (mostly) be in **R**
- This is industrial-strength, state-of-the-art, and free software for statistical computing
- We will access R through the **RStudio** graphical interface; make sure both are installed on your laptop and bring it to every class





Getting help

- My office hours
- Schedule an one-on-one time with me to chat (Zoom or in-person)

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The bias-variance tradeoff



What is supervised learning?

- Simply put, the goal is to **predict a target variable Y with input variables X**
- In data mining terminology, this is known as **supervised learning**
- We observe training data: $\{(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)\}$



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Tangent—what do you think **unsupervised learning** is?



The supervised learning framework

In general, a useful way to think about it is that Y and X are related in the following way:

$$Y_i = f(X_i) + \epsilon_i$$



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$$Y_i = f(X_i) + \epsilon_i$$

- $f(X_i)$ is the systematic part (the signal)
- ϵ_i is the random error (the noise)
- The main purpose is to *learn or estimate* $f(\cdot)$ from data



Two types of problems

1. **Regression:** When Y is quantitative (continuous)



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 - Examples: house prices, test scores, income, stock returns



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Today we'll focus on regression, but the same principles apply to classification



How do we estimate $f(\cdot)$?

1. Choose a set of **training data**: $(Y_1, X_1), \dots, (Y_n, X_n)$



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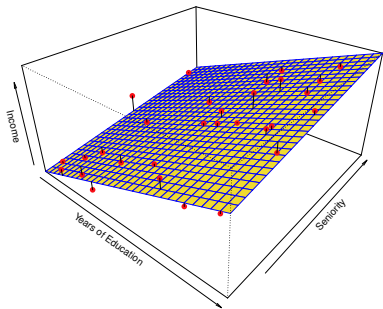
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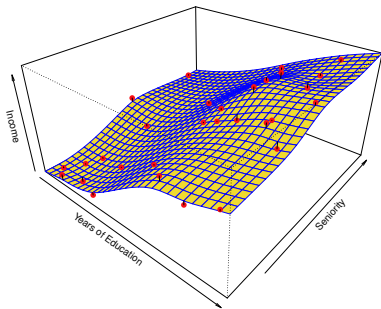
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 - **Nonparametric model**: Let the data determine the form of f (e.g., k-nearest neighbors)
3. Evaluate performance on **testing data** and *adjust*

Parametric vs. nonparametric



Parametric



Nonparametric

- **Parametric:** Restrictive assumptions, simple interpretation
- **Nonparametric:** Flexible assumptions, complex interpretation

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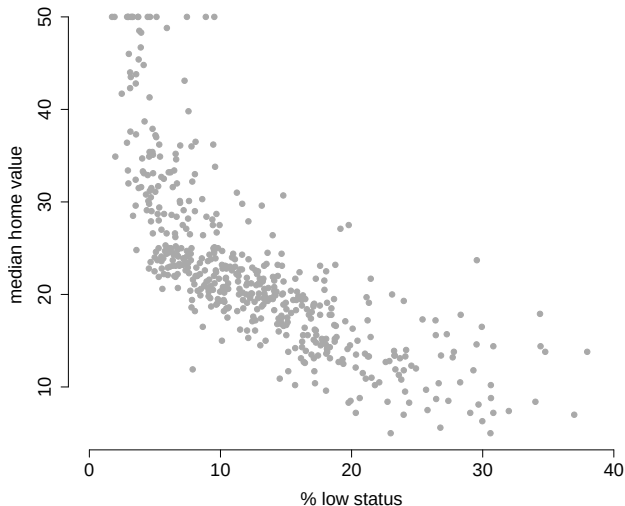
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Boston housing example

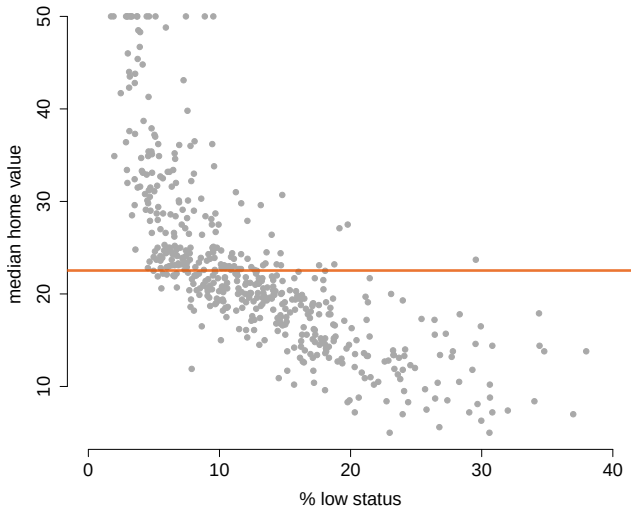
- Predict median home value with percent low economic status
- What should $f(\cdot)$ be?





What should $f(\cdot)$ be?

How about this? (Parametric: $Y = \mu + \epsilon$)

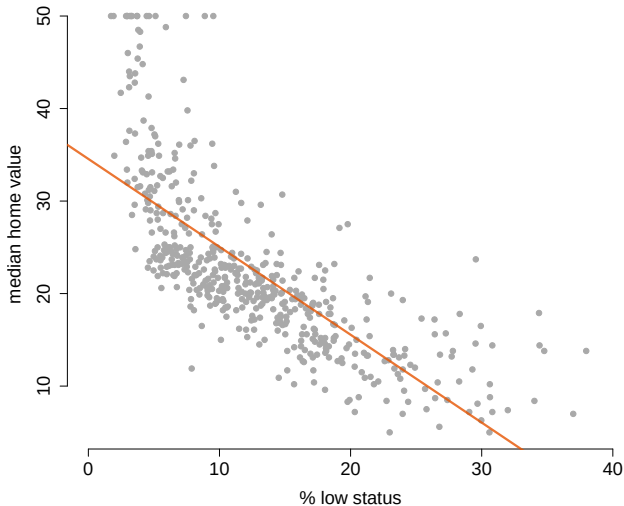


If % low status = 30, what is the prediction?



Or maybe this?

Linear model!

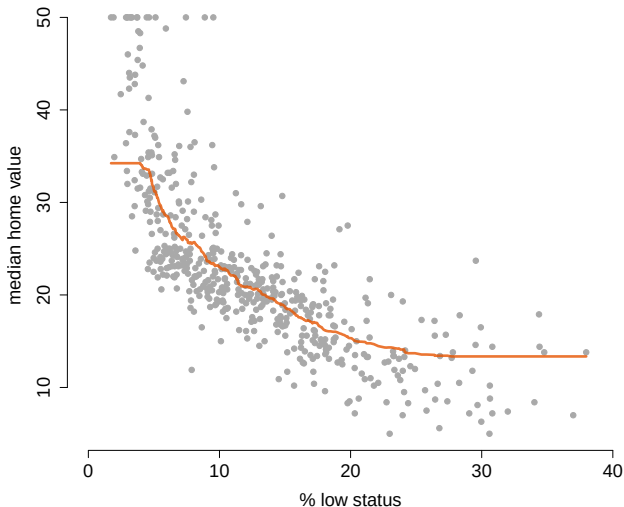


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Or even this?

(k -NN with $k = 100$)



If % low status = 30, what is the prediction?



k-nearest neighbors: How it works

Prediction at point x : $\hat{f}(x)$ = average of k nearest points around x



k-nearest neighbors: How it works

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- Points that are **closer** to where we want to predict should be more relevant
- We choose the k points closest to the X value we're trying to predict
- This is the **k -nearest neighbors** algorithm



k-nearest neighbors: How it works

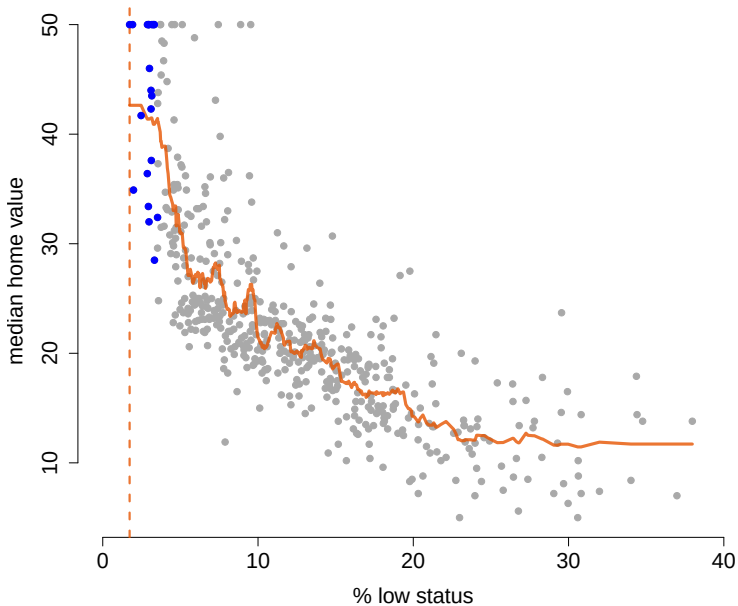
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Let's see how this works with $k = 20$...

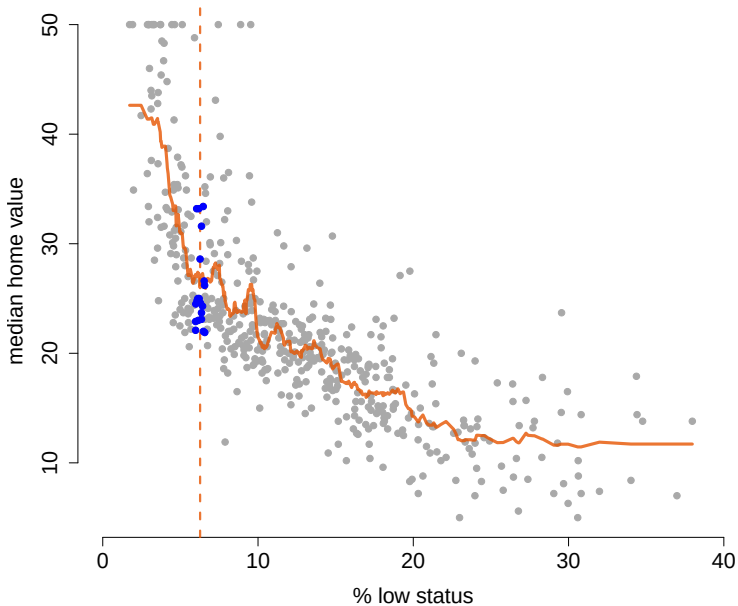


k-NN with $k = 20$



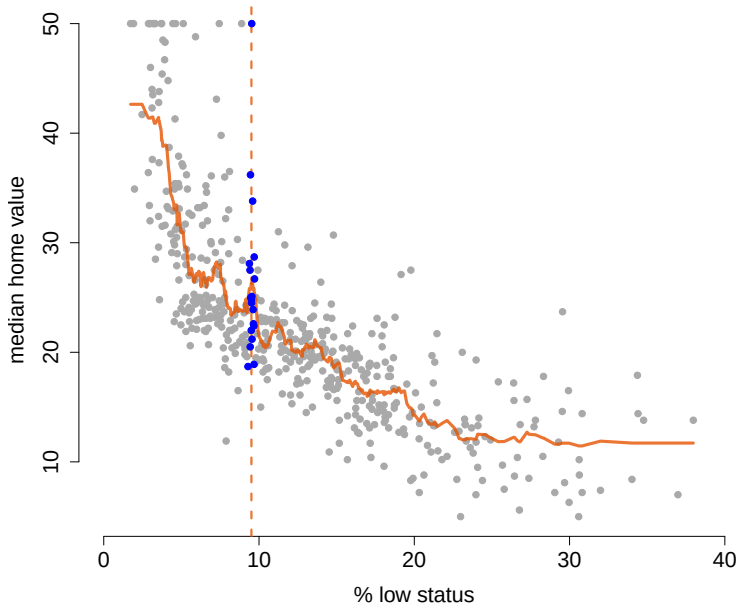


k-NN with $k = 20$



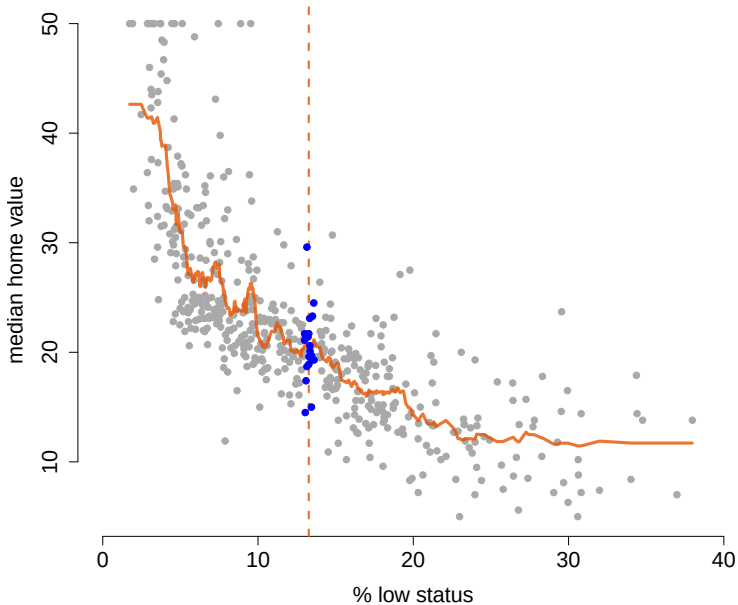


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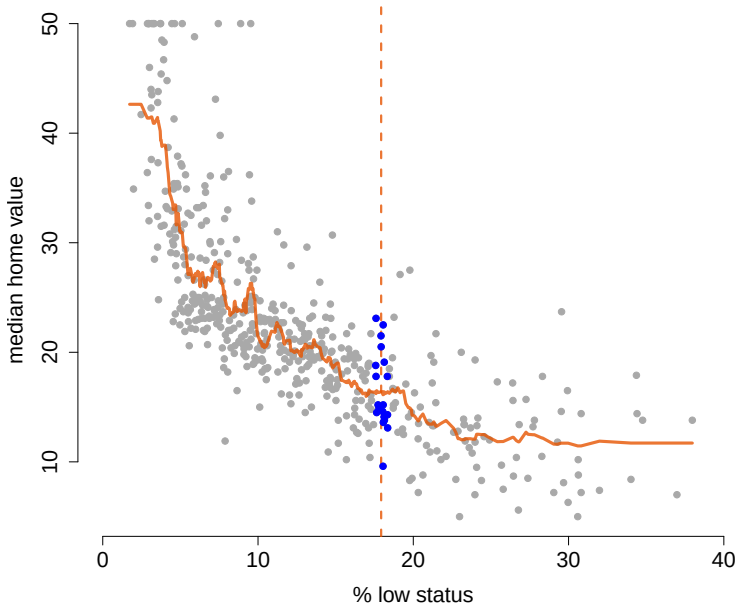


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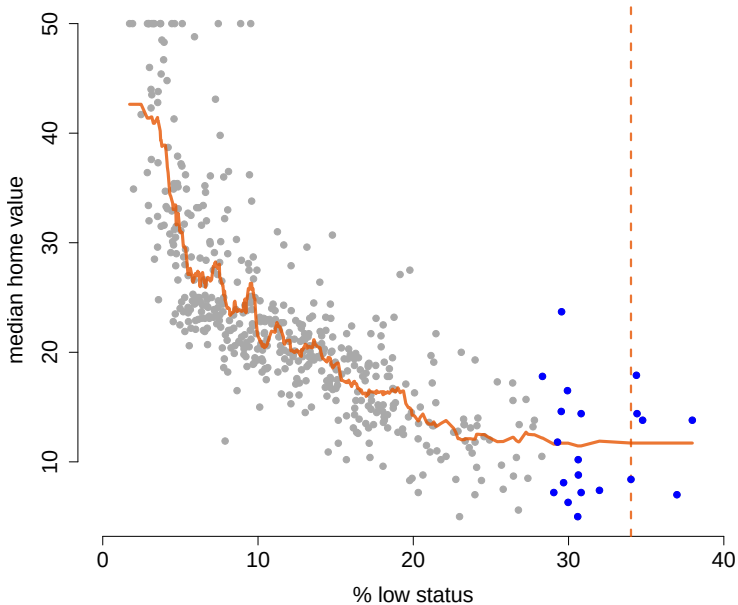




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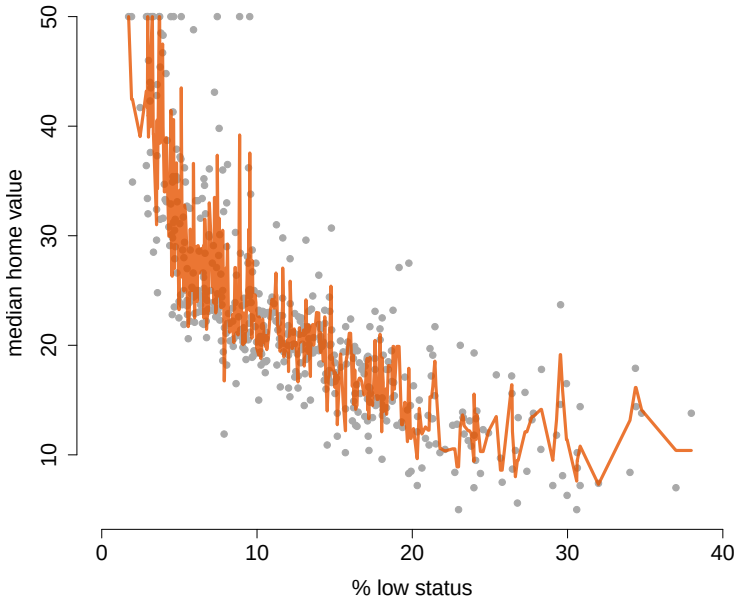


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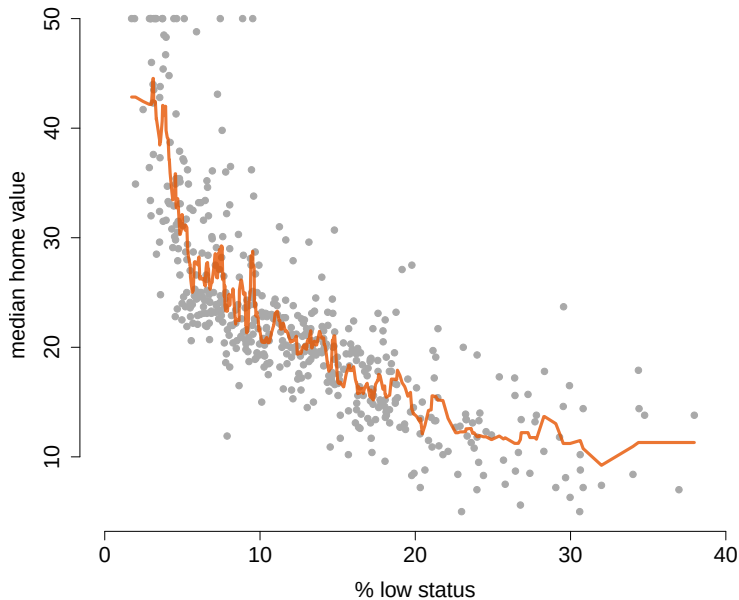


Why not choose $k = 2$?



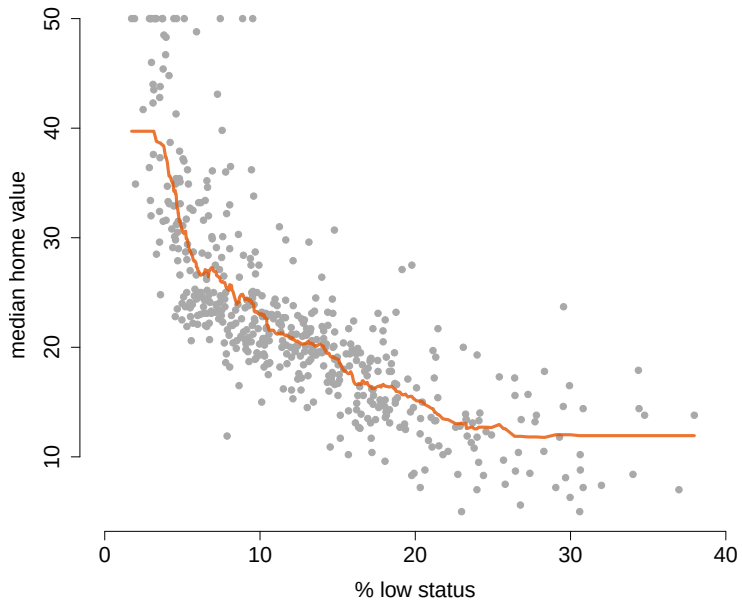


Or $k = 10$?



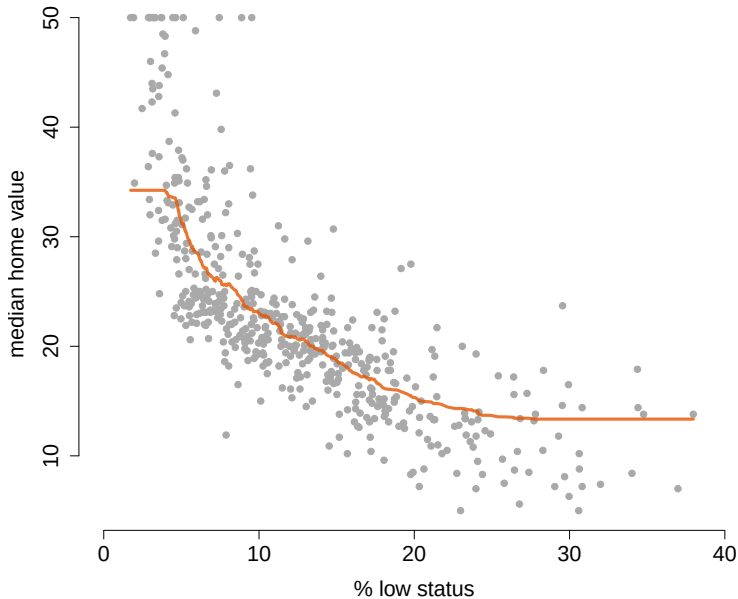


Or $k = 50$?



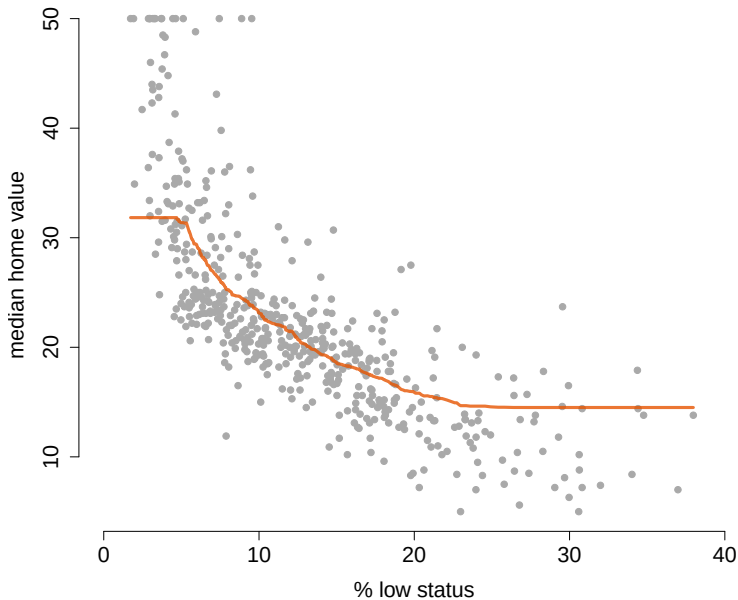


Or $k = 100$?



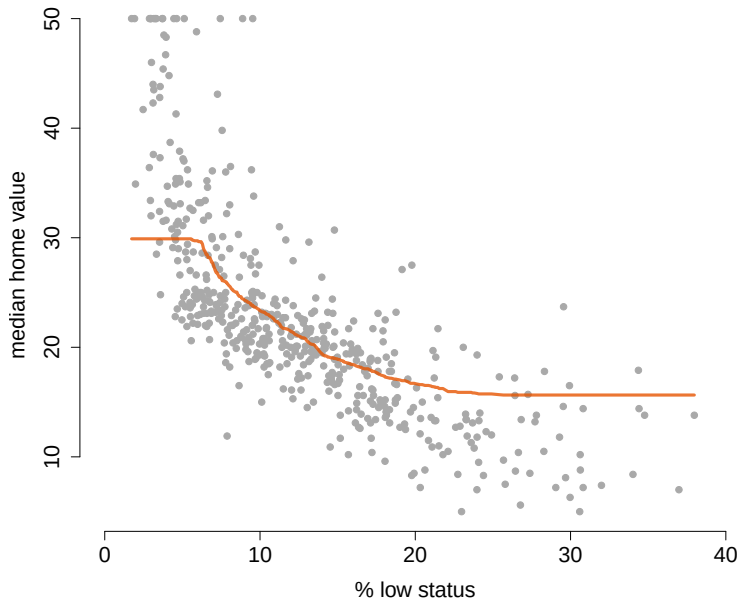


Or $k = 150$?



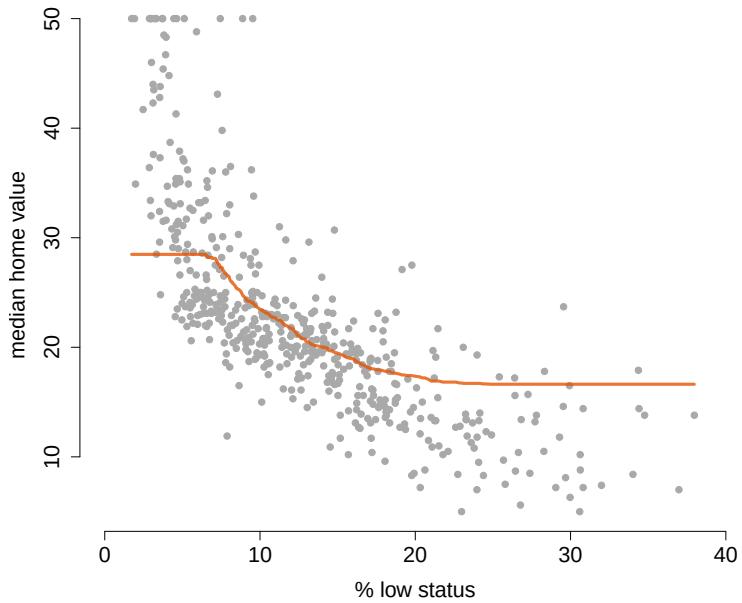


Or $k = 200$?



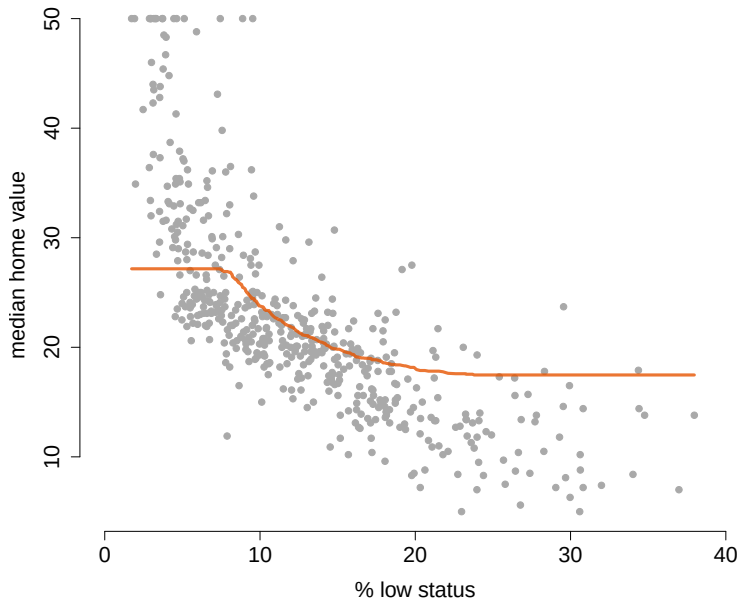


Or $k = 250$?



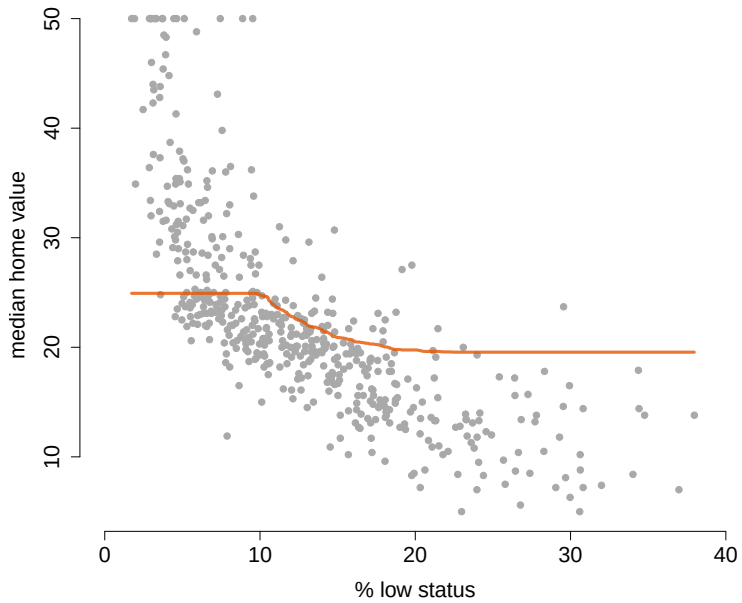


Or $k = 300$?



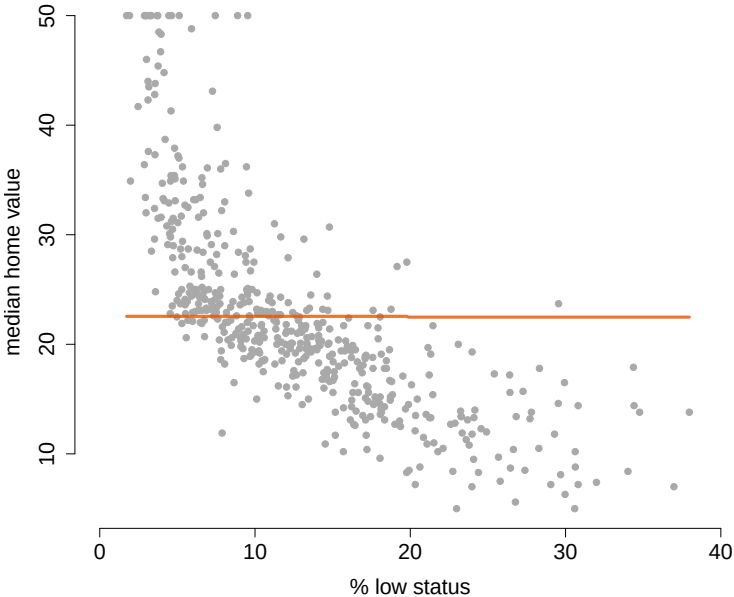


Or $k = 400$?





Or $k = 505$ (all data)?





Measuring prediction accuracy

- **In-sample error:** How well does the model fit the training data?

$$\text{RMSE}_{\text{in}} = \sqrt{\frac{1}{n} \sum_{i=1}^n [Y_i - \hat{f}(X_i)]^2}$$

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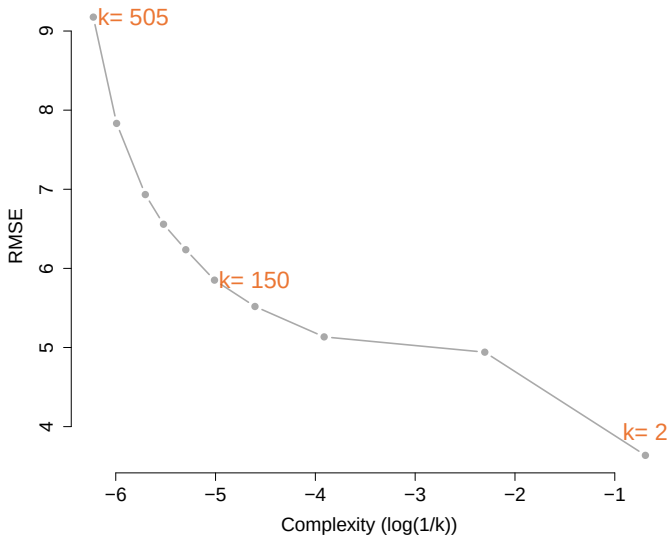
- **Out-of-sample error:** How well does the model predict new data?

$$\text{RMSE}_{\text{out}} = \sqrt{\frac{1}{m} \sum_{i=1}^m [Y_i^o - \hat{f}(X_i^o)]^2}$$

- We care about out-of-sample performance!



In-sample RMSE



It looks like $k = 2$ is the best. Should we choose this model?



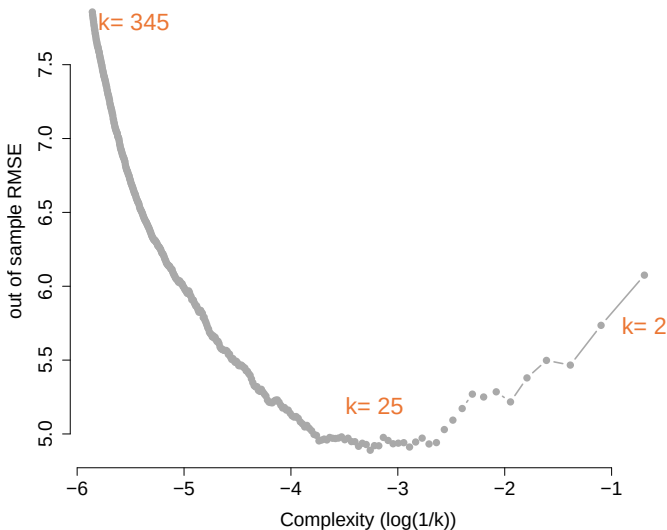
We care about out-of-sample performance!

- Suppose we have m additional observations (X_i^o, Y_i^o) that we **did not use** to fit the model
- This is called the **validation set** (or hold-out set, or test set)
- We evaluate with out-of-sample RMSE:

$$\text{RMSE}^o = \sqrt{\frac{1}{m} \sum_{i=1}^m [Y_i^o - \hat{f}(X_i^o)]^2}$$

Out-of-sample RMSE

Fit each model on training set (size 400). Test on validation set (size 106).

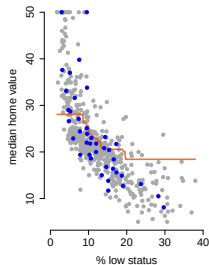
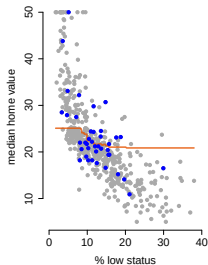
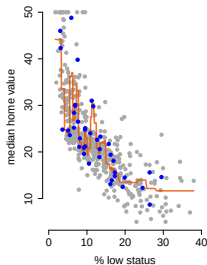
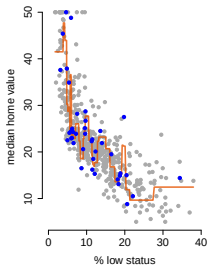




Visualizing bias and variance

$k = 2$: **Low bias, high variance**

$k = 25$: **High bias, low variance**



Training set of size 40

Training set of size 40

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A fundamental challenge

When fitting a predictive model, there is a tradeoff between:

- **Bias**: How far off our predictions are on average



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Our goal: Find the sweet spot in the bias-variance tradeoff!



Understanding the tradeoff

- **Simple models** (e.g., large k in k-NN):



Understanding the tradeoff

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 - May not capture all patterns → **high bias**



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 - May not capture all patterns → **high bias**
 - But predictions are stable → **low variance**
- **Complex models** (e.g., small k in k-NN):
 - Can capture complex patterns → **low bias**
 - But predictions vary a lot → **high variance**



The mathematical decomposition

For a prediction at point x , the mean squared error can be decomposed:

$$\text{MSE} = \text{E}[(f(x) - \hat{f}(x))^2] = \underbrace{[\text{E}[\hat{f}(x)] - f(x)]^2}_{\text{Bias}^2} + \underbrace{\text{Var}[\hat{f}(x)]}_{\text{Variance}}$$



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- **Bias²**: How far off our average prediction is from the truth



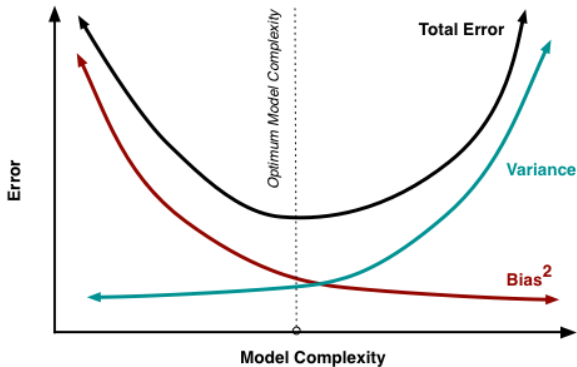
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- **Bias²**: How far off our average prediction is from the truth
- **Variance**: How much our predictions vary

Visualizing the tradeoff



- As model complexity increases, bias decreases but variance increases
- The optimal model complexity minimizes the total error



Why out-of-sample matters

- A model that fits the training data perfectly may not generalize to new data



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- This is called **overfitting**
- We want models that:
 - Capture important patterns that generalize
 - Don't chase noise in the training data
- **The bias-variance tradeoff helps us understand this balance**



Key takeaways

1. **Supervised learning** is about predicting Y from X by learning $f(\cdot)$



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2. We can use **parametric** or **nonparametric** methods



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3. There's a fundamental **bias-variance tradeoff**:



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1. **Supervised learning** is about predicting Y from X by learning $f(\cdot)$
2. We can use **parametric** or **nonparametric** methods
3. There's a fundamental **bias-variance tradeoff**:
 - Simple models: high bias, low variance



1. **Supervised learning** is about predicting Y from X by learning $f(\cdot)$
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- The bias-variance tradeoff will continue to guide our thinking throughout the course!

Extra Slides



Deriving the decomposition

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Squaring both sides gives squared terms **plus cross terms**:

$$(Y - \hat{f}(X))^2 = [f(X) - E[\hat{f}(X)]]^2 + [E[\hat{f}(X)] - \hat{f}(X)]^2 + \epsilon^2 \\ + (\text{cross terms})$$



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For the MSE $E[(f(X) - \hat{f}(X))^2]$, the cross terms vanish, leaving only the two squared terms: bias^2 and variance.